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COURSE CODE :CSA0920

COURSE NAME : PROGRAMMING IN JAVA

ASSIGNMENT : ASSIGNMENT – 2

QUESTION : Q4

QUESTION

Q4. Multi-Catch Statement (Single Catch, Multiple Types)

Write a Java program that demonstrates Java's multi-catch feature, where a single catch block can handle more than one exception type using the pipe (|) syntax. Trigger a scenario where either a `NumberFormatException` or `ArithmeticException` could occur (such as parsing user input into a number and then dividing it), and catch both exception types in one combined catch block.

GitHub Repository Link: <https://github.com/vijayy1234-web/CSA0920-PROGRAMMING-IN-JAVA.git>

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2. PROBLEM UNDERSTANDING

The objective of this program is to demonstrate Java's **multi-catch** feature. It performs an operation that may throw either a `NumberFormatException` or an `ArithmeticException` and handles both exceptions using a single catch block.

3. APPROACH / DESIGN NOTES

- Used `Integer.parseInt()` to convert a string into an integer.
- Performed a division operation that may throw an exception.
- Used a single multi-catch block with the `|` operator.

- Printed the exception message when an exception occurred.
- This approach reduces duplicate exception handling code.

4.SOURCE CODE

```
public class MultiCatchDemo {
    public static void main(String[] args) {

        String input = "10";
        int divisor = 0;

        try {
            int number = Integer.parseInt(input);
            int result = number / divisor;

            System.out.println("Result: " + result);

        } catch (NumberFormatException | ArithmeticException e) {
            System.out.println("Exception caught: " + e);
        }
    }
}
```

5.TEST CASES

TEST CASE	INPUT	EXPECTED OUTPUT	ACTUAL OUTPUT
1 (Normal)	input = "10", divisor = 2	Result: 5	Result: 5
2 (Edge case)	input = "10", divisor = 0	Exception caught: java.lang.ArithmeticException: / by zero	Same as Expected
3(Boundary Case)	input = "abc", divisor = 2	Exception caught: java.lang.NumberFormatException	Same as Expected

6. SCREENSHOTS OF EXECUTION

Insert your execution screenshot here.

Your output for the uploaded program is:

```
Output
Exception caught: java.lang.ArithmeticException: / by zero

=== Code Execution Successful ===
```

7. REFLECTION

I learned how to use Java's **multi-catch** feature to handle multiple exception types with a single catch block. This makes exception handling simpler, reduces code duplication, and improves program readability.

8. DECLARATION

This is my own work. No AI/external tool assistance was used except for formatting and documentation preparation.